IEEE CyberSciTech / DASC / PICom / CBDCom 2026

Detailed Program · 9 to 13 November 2026 · RMIT University, Building 80, Melbourne · All times AEDT (UTC+11)

Messages from the Chairs

Message from CyberSciTech/DASC/PICom/CBDCom 2026

Steering Committee Chairs

Welcome to the four joint IEEE conferences of CyberSciTech, DASC, PICom and CBDCom! It is very delightful for us to gather together, physically and virtually, in Melbourne, a historic and beautiful city in Australia, in the late spring season.

This year our joint conferences have received a large number of submissions from all over the world. After rigorous review, the four conference program committees have selected a total of *201* papers, including *126* regular ones, *13* work-in-progress or short papers, *12* symposium papers, and *50* workshop or special session papers. In addition, there are *three* panel sessions covering various hot topics and one panel for discussing the future of AI.

It is very challenging to organize such a scale of the joint conferences. There have been a lot of organizing tasks since the beginning of this year. Many people have spent tremendous time and done excellent work to make the joint event successful. First of all, we would like to thank RMIT University, Prof. *Yang Xiang* and Prof. *Chao Chen*, for hosting the conferences so that we all can meet here. Next, we would like to express our gratitude to the conferences' leading general and program chairs, especially, *Feng Xia, Wenny Rahayu, Xiaokang Zhou, Robert Deng, Willy Susilo, Yang Xiang, Claudio Miceli, Eric Pardede, Boualem Benatallah, David Taniar, Xun Shao, Celimuge Wu, Chao Chen, Bernady O. Apduhan, Sahraoui Dhelim, Chaofeng Zhang, Ao Guo, Fengling Han, Huan Nguyen, Weizhi Meng, Wei Xiang, Rossana Andrade, Bo Wu, Cheah Wai Shiang, Kiki Adhinugraha* and *Siyang Lu*.

We highly appreciate the prestigious forum chairs, Prof. *Qiangfu Zhao* and Prof. *Yang Xiang*, for organizing FFRise, Forum for Future Research, Innovation and Education (Panels R, I and E in this program), as well as the panel chairs, Prof. *Chao Chen*, Prof. *Xingliang Yuan*, Prof. *Qing Li*, Prof. *Zhong Chen*, Prof. *Wei Lu*, Prof. *Xiao Huang*, and all the distinguished panelists. We also appreciate the HAI-Q, Hyper/Super AI Questions Initiative, open forum organizers, Prof. *Qun Jin*, Prof. *Yang Xiang*, Prof. *Chao Chen*, Dr. *Ao Guo*, Prof. *Binbin Zhou*, Dr. *Ruichen Cong* and Ms. *Lifei Wang*.

We also sincerely thank the advice, support and excellence of all other chairs. In particular, our special thanks go to Prof. *Chao Chen*, who has led and managed all technical and local matters, as well as to Dr. *Wanlun Ma*, who has done tremendous work in maintaining the multi-conference websites, systems, programs, etc, and to Dr. *Ruiqi Dong*, who has provided a lot of detailed local support. We also appreciate the excellent service of the volunteers, who have done a lot of detailed support but often invisible work, e.g., logistics preparation, registration administration, facility preparation and setting up, etc.

Finally, we are grateful to all authors submitting high-quality research works to our conferences, and all participants attending the four conferences.

We hope you will fully enjoy both the joint conferences and Melbourne, Australia.

Jianhua Ma, Hosei University, Japan
Steering Committee Chair

Laurence T. Yang, St. Francis Xavier University, Canada
Steering Committee Chair

Message from CyberSciTech 2026

General Chairs, General Executive Chairs and Program Chairs

Over the past decade, cyber science and technology is developing at a tremendous rate. It is becoming an inseparable part of our daily lives. At the same time, challenges across cyber technologies, ethics, and other traditional disciplines are emerging. The future of this theme will be of transdisciplinary nature that requires careful exploration. The aim of the CyberSciTech Congress is to address the broad challenges in cyber science and technology and to offer a common platform for our fellow scientists, engineers, industrial practitioners, and researchers to present and exchange their latest ideas, discoveries, and implementations. Therefore, it is our great honor and pleasure to welcome all our participants to the 2026 Cyber Science and Technology Congress (CyberSciTech 2026) to be held in Melbourne, Australia during November 9-13, 2026. CyberSciTech 2026 is sponsored by the IEEE Computer Society, supported by IEEE Technical Committee on Cybermatics, IEEE Technical Committee on Smart World, IEEE Technical Committee on Scalable Computing, and IEEE Technical Committee on Hyper Intelligence, and hosted by RMIT University, Australia. It is co-located with the 24th IEEE International Conference on Dependable, Autonomic & Secure Computing (DASC 2026), the 24th IEEE International Conference on Pervasive Intelligence and Computing (PICom 2026), and the 12th IEEE International Conference on Cloud and Big Data Computing (CBDCom 2026).

To address the comprehensive nature and emerging challenges of Cyberization, CyberSciTech 2026 offers six technical tracks on the topics of

- Track 1: Cyberspace Theory & Technology
- Track 2: Cyber Security, Privacy & Trust
- Track 3: Cyber Physical Computing & Systems
- Track 4: Cyber Social Computing & Networks
- Track 5: Cyber Intelligence & Cognitive Science
- Track 6: Cyber Life & Wellbeing

In addition, the following symposium/special sessions/workshops are jointly organized:

1. The 3rd International Symposium on Generative AI and Hyper Intelligence (GAI-HyperI 2026)
2. Special Session on Large Model Empowered Cyber Social Computing and Cyber-Enabled Applications (LM-CSC&CEA 2026)
3. Special Session on Security and Trustworthiness of Large AI Models and Agents (SecTrust-AI 2026)
4. Special Session on Computing and Applications for Cyber Internet of Things (Cyber-IoT 2026)
5. The 2nd International Workshop on AI-empowered Digital Health and Well-being Promotion (AI-DHWP 2026)
6. The 9th International Workshop on Applications of AI, Cyber Security and Economics Big Data (ACE-2026)
7. The 10th International Workshop on the Impact of Internet of Things on Daily Life (IoT-Life 2026)
8. International Workshop on Human-Centered Multimodal and Generative Intelligence for Cyber-Physical-Social Systems (HMGI-CPSS 2026)
9. The 2nd International Workshop on Human-centered AI and Digital Coaching (HAIDC 2026)

Overall, CyberSciTech 2026 received 201 submissions covering a wide range of topics. All accepted papers are selected based on a rigorous peer review process. Finally, 56 high-quality regular papers, 7 Work-in-Progress papers, 12 symposium papers and 38 special sessions/workshops papers are included in the 2026 Proceedings.

We would like to take this chance to thank the entire Steering Committee, especially Jianhua Ma (Chair), Qun Jin, Laurence Yang, and Hui-Huang Hsu for their guidance.

It is our great pleasure to find and assemble a great organizing committee, which is the key to a successful event. We want to thank RMIT University and all members of the Organizing Committee for their hard work. We would also like to express our gratitude to the Program Co-Chairs, Huaming Chen, Yan Huang, Xiaoyan Wang, Weimin Li, S. Leili Mirtaheri, Lai Tu, Chen Yang, Qin Wang and Binbin Zhou, for their valuable contributions.

We thank workshop and special session chairs Pan Wang, Yegang Du and Xiwei (Sherry) Xu in inviting and organizing many great and topical workshops and special sessions to enrich the coverage of our Congress. A successful event always relies on a great publicity team and special thanks must go to Xiaohua Feng and Hongxin Yan.

Of course, we want to express our sincere gratitude to all authors, participants, PC members, and many others who greatly contributed to CyberSciTech 2026 in many ways.

The event will take place in Melbourne, Australia. We are excited to welcome our global community and present the exceptional papers and presentations prepared by our colleagues worldwide. We hope this conference will offer valuable insights and inspiration, fostering meaningful exchanges and collaborations. We look forward to your participation and wish you an enjoyable and productive experience at CyberSciTech 2026.

Feng Xia, RMIT University, Australia
Wenny Rahayu, La Trobe University, Australia
Xiaokang Zhou, Kansai University, Japan
General Chairs of CyberSciTech 2026

Celimuge Wu, The Univ. of Electro-Comm., Japan
Chao Chen, RMIT University, Australia
General Executive Chairs of CyberSciTech 2026

Ao Guo, Nagoya University, Japan
Fengling Han, RMIT University, Australia
Huan Nguyen, Middlesex University, UK
Program Chairs of CyberSciTech 2026

Message from DASC 2026

General Chairs, General Executive Chairs and Program Chairs

As computer and communication systems continue to evolve in scale, complexity, and intelligence, dependability, security, privacy, and trust have become increasingly essential for enabling next-generation scientific, engineering, and commercial applications. The rapid development of Cyber-Physical Systems (CPS), the Internet of Things (IoT), autonomous robotic systems, and AI-enabled computing has created new opportunities while also introducing significant challenges in designing, analyzing, evaluating, and enhancing the dependability and security of increasingly complex computing environments. Achieving trustworthy, dependable, secure, and autonomic computing systems requires close collaboration across multiple disciplines. This includes advances in software systems, computing architectures, communication technologies, artificial intelligence, data platforms, cybersecurity, and autonomous systems. In particular, the rapid emergence of large language models, generative AI, embodied AI, and other advanced computational technologies is creating new research opportunities for building intelligent systems that are not only capable, but also dependable, secure, explainable, and trustworthy. It is therefore our great honor and pleasure to welcome all participants to the 24th IEEE International Conference on Dependable, Autonomic and Secure Computing (DASC 2026), which will be held in the vibrant city of Melbourne, Australia, during November 9–13, 2026. DASC 2026 is sponsored by the IEEE, supported by IEEE Technical Committee on Cybermatics, IEEE Computer Society, IEEE Technical Committee on Scalable Computing, IEEE Technical Committee on Hyper Intelligence. DASC 2026 is co-located with the 11th IEEE Cyber Science & Technology Congress (CyberSciTech 2026), the 24th IEEE International Conference on Pervasive Intelligence & Computing (PICom 2026), and the 12th IEEE International Conference on Cloud & Big Data Computing (CBDCom 2026).

To address the broad and rapidly evolving challenges in dependable, autonomic, and secure computing, DASC 2026 features five technical tracks covering a wide range of emerging research topics:

- Track 1: Dependability, Reliability, and Fault Tolerance
- Track 2: Security, Privacy, and Trust in Systems and Networks
- Track 3: Autonomic Computing and Self-Adaptive Systems
- Track 4: AI, Machine Learning, and Advanced Computational Methods
- Track 5: Industrial Applications and Case Studies

In addition, the following two workshops on the topics of

1. The 10th International Workshop on Emerging Dependable Computing System Technologies and Applications (EDCSTA 2026)
2. The 24th International Workshop on Assurance in Distributed Systems and Networks (ADSN 2026)

are jointly organized.

Overall, DASC 2026 attracted high-quality submissions from various regions worldwide, encompassing a broad spectrum of topics. Finally, 19 high-quality regular papers were accepted for presentation and publication in the conference proceedings. As the number of submitted papers to main tracks (1-to-5) and the Late Breaking Innovation track is 44, the acceptance rate is 43%. Besides the main tracks, 1 Work-in-Progress (WiP) paper was accepted, together with 8 papers in the two co-located workshops.

We would like to take this opportunity to express our sincere appreciation to the DASC Steering Committee and all members of the organizing committee for their guidance and support. We are especially grateful to the Honorary Chairs Wanlei Zhou and Elisa Bertino, and Advisory Committee Naohiro Hayashibara, Takaki Nakamura, and Zonghua Zhang whose dedication and hard work have contributed to the organization of DASC 2026.

We would also like to express our sincere gratitude to Ao Guo, Ricky Dong, Fuyi Wang, Di Wu, Wanlun Ma, and Xiaogang Zhu whose dedication and hard work have contributed to the organization of DASC 2026. We would also like to express our gratitude to the reviewers for their valuable contributions. We thank Workshop and Special Session Chairs Viet Vo and Leo Yu Zhang in organizing great and topical workshops, WiP, LBI, poster and special sessions to enrich the coverage of our Congress. A successful event always relies on a great publicity team and special thanks must go to Viet Vo, Leo Zhang, Chaoqi Jia, Ruichen Cong, Lin Li, and Jiaheng Wei. Most importantly, we thank all authors and participants who contributed to DASC 2026 by submitting their work, sharing their research, participating in technical discussions, and engaging with colleagues from around the world. Their contributions are at the heart of the conference and help make DASC an active and vibrant international research community.

The event will be held on the beautiful city Melbourne, Australia. We are thrilled to welcome our global community and showcase the outstanding papers and presentations from our colleagues around the world. We hope this conference provides valuable insights and inspiration, encouraging meaningful discussions and collaborations. We look forward to your participation and wish you an enjoyable and fruitful experience at DASC 2026.

Robert Deng, Singapore Management University, Singapore
Willy Susilo, University of Wollongong, Australia
Yang Xiang, Monash University, Australia
General Chairs of DASC 2026

Chao Chen, RMIT University, Australia
Bernady O. Apduhan, Kyushu Sangyo University, Japan
General Executive Chairs of DASC 2026

Weizhi Meng, Lancaster University, UK
Wei Xiang, La Trobe University, Australia
Chao Chen, RMIT University, Australia
Program Chairs of DASC 2026

Message from PICom 2026

Program Chairs and General Chairs

On behalf of the Organizing Committee, we welcome you to the 24th IEEE International Conference on Pervasive Intelligence and Computing (PICom 2026). Since its first edition in 2003, PICom has been established as a premier conference aiming to cover all dimensions of intelligent paradigms and their applications in several pervasive computing domains. By covering the different visions and components of computational intelligence, including well established techniques and disciplines as well as emerging paradigms, PICom has a broad scope. It aims to provide a forum to discuss and exchange novel and exciting ideas in the wide spectrum of theoretical and applied pervasive and intelligent computing. PICom's scope includes Ubiquitous Intelligence, Social Intelligence, Intelligent Networks, Machine Learning, Big Data, Internet of Things, Cloud Computing, Context-Aware Computing, and Pervasive Security, to name a few.

This year, we received high-quality submissions from several parts of the world covering a wide range of topics. A total of 16 Full Research papers and 5 work-in-progress papers were accepted for presentation and publication in the conference proceedings. Besides the main tracks, PICom also included one very interesting workshop, the 2nd International Workshop on Quantum Computing for Distributed Systems (QuDiS 2026).

The successful organization of an international conference of this scale requires the support and dedication of many people. First, we would like to thank the Steering Committee, Prof. Jianhua Ma, Prof. Laurence T. Yang, and Prof. Adnan Al-Anbuky, for nourishing the conference and wisely guiding its course over the years. A warm thanks to Prof. Bernady O. Apduhan, who (again) accepted the challenge of organizing the event and making everything run smoothly and efficiently. In addition to meeting all the numerous requests involved in the organization, Prof. Bob Apduhan always does so with a smile on his face. We extend our thanks to the entire local committee members for their excellent and diligent work. We also want to thank Xiaokang Zhou and Ao Guo for their very important contribution to the conference organization. We are indebted to our Track Chairs - Santoso Wibowo, Lidia Fotia, Gautam Srivastava, Alex Tomy, Rebeca Motta and Marco Miozzo - for efficiently guiding the review process with the invaluable help of the members of the Technical Program Committee. Without their help, this program would not be possible. We are sincerely grateful to the Workshop & Special Session Chairs - Victor Chang and Federico Santoro - and to our Publicity Chairs - Antonio Guerrieri, Rudri Kalaria, Sun Jingtao and Hong Chen - for helping in the dissemination of the Call for Papers and the members of the Advisory Committee.

A heartfelt thanks to all the authors for their great contributions. We are also grateful to IEEE for publishing the proceedings. We sincerely hope the participants enjoy the conference program and benefit from all the learning opportunities and academic and professional interactions.

The 2026 edition of PICom is held in the beautiful Melbourne, Australia. We hope that participants truly enjoy the numerous local attractions, the hospitality of its people and the great quality of life.

Claudio Miceli, Federal University of Rio de Janeiro, Brazil
Eric Pardede, La Trobe University, Australia
General Chairs of PICom 2026

Rossana Andrade, Federal University of Ceara, Brazil
Bo Wu, Tokyo University of Technology, Japan
Program Chairs of PICom 2026

Message from CBDCom 2026

General Chairs, General Executive Chairs and Program Chairs

On behalf of the Organizing Committee, it is our great honor and pleasure to welcome you to the 12th IEEE International Conference on Cloud and Big Data Computing (CBDCom 2026), held from November 9-13, 2026, in Melbourne, Australia, with hybrid participation options. Since its inception, CBDCom has established itself as a premier international forum for presenting the latest research innovations, systems, and applications in cloud computing and big data technologies. Over the past decade, the conference has consistently provided a unique venue for researchers, practitioners, and industry leaders to exchange ideas and foster collaborations that shape the future of computing.

The significance of cloud and big data computing continues to grow. The exponential rise of data generated by the Internet of Things, smart devices, social media, and large-scale scientific applications demands new paradigms for data storage, processing, and analysis. At the same time, the increasing adoption of AI and machine learning techniques requires scalable, secure, and energy-efficient infrastructures. Addressing these challenges calls for innovative approaches that combine advanced algorithms, novel architectures, distributed systems, and networked intelligence. CBDCom 2026 is designed to provide exactly this type of comprehensive platform. This year's conference features four technical tracks covering theories and algorithms, systems and security, IoT and smart technologies, and networking and communications. These tracks reflect the multi-disciplinary nature of the field and ensure that diverse perspectives are represented.

This year, the conference received a substantial number of submissions from around the world. We would like to thank the Program Chairs for their tireless efforts in coordinating the review process and shaping the program. We are equally grateful to the Workshop and Special Session Chairs, Publicity Chairs, and Publication Chair, whose contributions have been essential to the organization of the conference. Our sincere thanks also go to the Steering and Advisory Committees for their invaluable guidance, as well as to IEEE for publishing the proceedings. We also acknowledge the contributions of our local hosts and volunteers in Melbourne, whose hard work has made this event possible. Melbourne, located in southeastern Australia, is renowned for its scenic beauty, historical sites, and vibrant culture. We hope that in addition to the rich technical program, participants will enjoy exploring this unique city and engaging in cultural exchange.

Finally, we warmly welcome all participants to CBDCom 2026, whether you are joining us in person in Melbourne or virtually from around the world. We hope this conference will provide not only technical insights but also opportunities to establish long-lasting collaborations and friendships. We wish you an enjoyable, fruitful, and memorable experience at IEEE CBDCom 2026.

Boualem Benatallah, Dublin City University, Ireland
David Taniar, Monash University, Australia
Xun Shao, Toyohashi University of Technology, Japan
General Chairs of IEEE CBDCom 2026

Chao Chen, RMIT University, Australia
Sahraoui Dhelim, Dublin City University, Ireland
Chaofeng Zhang, Tokyo University of Technology, Japan
General Executive Chairs of IEEE CBDCom 2026

Cheah Wai Shiang, Universiti Malaysia Sarawak, Malaysia
Kiki Adhinugraha, La Trobe University, Australia
Siyang Lu, Beijing Jiaotong University, China
Program Chairs of IEEE CBDCom 2026

Message from GAI-HyperI 2026

General Chair and Program Chair

Generative AI and Hyper Intelligence are rapidly converging to enable more capable, adaptive, and human-centered intelligent systems. To foster research and discussion at this intersection, GAI-HyperI was launched as an international workshop in 2024, with its first two editions held in conjunction with IEEE CyberSciTech 2024 in Boracay Island, Philippines, and IEEE CyberSciTech 2025 in Hakodate, Japan. Building on the growing research community and expanding scope of the field, GAI-HyperI has been elevated to an **international symposium** in 2026. We are delighted to welcome you to the **3rd International Symposium on Generative AI and Hyper Intelligence (GAI-HyperI 2026)**, held in conjunction with IEEE CyberSciTech 2026 in Melbourne, Australia.

The theme of GAI-HyperI 2026 is **"Beyond Generation."** This theme reflects the ongoing transition of Generative AI from generating content toward systems capable of reasoning, planning, using tools, maintaining memory, interacting with the physical world, collaborating with humans and other agents, and continuously adapting to changing environments. The symposium covers a broad spectrum of topics ranging from foundation, multimodal, and reasoning models to agentic AI, world models, embodied intelligence, human-AI collaboration, collective intelligence, hyper-connected intelligent systems, as well as the safety, governance, and societal implications of increasingly capable AI.

This year, GAI-HyperI 2026 received **26 submissions** covering a diverse range of emerging topics in Generative AI, Hyper Intelligence, and their convergence. Following a rigorous peer-review process, **12 papers were accepted**. We believe that these selected contributions represent timely and promising advances in this rapidly developing area.

In addition to the technical paper presentations, GAI-HyperI 2026 introduces the **HAI-Q: Hyper/Super AI Questions Initiative**, which invites the community to identify and discuss fundamental and forward-looking questions concerning the future of Hyper and Super AI. Selected questions will form the basis of a dedicated symposium panel, providing an opportunity to look beyond individual technical advances and collectively discuss the long-term directions, challenges, and opportunities of future intelligent systems.

GAI-HyperI 2026 also features a **hands-on tutorial jointly organized with IEEE CyberSciTech 2026**, entitled *"Full-Duplex Spoken Dialogue for Embodied AI: Applications in Virtual Avatars and Humanoid Robot Simulation."* The tutorial introduces recent developments in spoken dialogue systems, with particular emphasis on full-duplex interaction, and provides practical experience in connecting dialogue systems with virtual avatars and simulated humanoid robots.

We would like to express our sincere gratitude to all contributors to the **HAI-Q: Hyper/Super AI Questions Initiative** and to the panelists for sharing their perspectives on the fundamental questions shaping the future of Hyper and Super AI. We sincerely appreciate all authors for contributing their latest research and ideas to GAI-HyperI 2026, and all participants for joining us and enriching the symposium through their discussions and exchanges.

We hope that GAI-HyperI 2026 will provide an inspiring forum for exchanging ideas, discovering new research directions, and establishing meaningful collaborations across disciplines. We look forward to welcoming you to Melbourne and wish everyone a stimulating, enjoyable, and productive symposium.

Jianhua Ma, Hosei University, Japan
General Chair of GAI-HyperI 2026

Ao Guo, Nagoya University, Japan
Program Chair of GAI-HyperI 2026

General Information and Presentation Guidelines

Conference Time and Language

All times in this program are Australian Eastern Daylight Time (AEDT, UTC+11), Melbourne time. The presentation language is English.

Venue and Registration

All sessions take place at RMIT University, Building 80, Melbourne. Registration, morning and afternoon tea and lunch are on Level 10. The Welcome Reception on Monday 9 November at 18:00 is in 080.07.005 on the Level 7 balcony.

Registration opens at 08:00 on Monday 9 November and at 08:30 from Tuesday 10 to Friday 13 November.

Rooms

080.01.002 on Level 1 for the Opening, keynotes, Panels I, R and Q, BoF and Closing. 080.10.012, 080.10.013, 080.10.014 and 080.10.016 on Level 10 for the sessions and the tutorials, with Panel E in 080.10.013. 080.09.012 on Level 9 L9 for online-only sessions. A room number reads building, level and room, so 080.10.013 is Building 80, Level 10, room 013.

Reading This Program

Session codes are CST = CyberSciTech, DASC, PICOM = PICom, CBD = CBDCom and TUT = Tutorial. Types are Regular (regular papers), Wksp (workshop or symposium), SS (special session) and WiP (work in progress), as in the Program at a Glance. Online sessions have every talk online. In Hybrid sessions, online marks a talk given over Zoom.

For Session Chairs

- Please be in the room at least 10 minutes before your session and check which presenters are in the room or online.
- Depending on the number of papers and the presenters' attendance, you can adjust the Q&A time, but please make sure the session keeps to its allocated time.
- For an online talk, please make sure the presenter has joined the room's Zoom meeting before the session starts.

For Authors

- Please confirm your attendance with the session chair before the session.
- Please check this program for the exact time of your session and the order of the talks in it.
- You are encouraged to use your own laptop for your presentation.
- Regular papers have 20 minutes, 15 for the presentation and 5 for Q&A.
- Workshop, symposium and special session papers have 15 minutes, 12 for the presentation and 3 for Q&A.
- Work-in-Progress papers have 10 minutes, 8 for the presentation and 2 for Q&A.

For Online Presenters

- Please join the Zoom meeting of your session's room at least 10 minutes before the session starts, with your name and paper ID as your display name.
- Please share your own screen when the session chair introduces your talk.

Zoom Meetings

Each session room has one Zoom meeting. Online presenters and participants join the meeting of the session's room with the meeting ID and passcode below.

Room	Used for	Meeting ID	Passcode
080.10.012	PICom and DASC sessions, and the online-only DASC-01, PICOM-01 and CBD-08 in 080.09.012	**846 7492 1018**	**664387**
080.10.013	CyberSciTech sessions and Panel E, and the keynotes and Panels I, R and Q in 080.01.002	**961 9676 7132**	**960648**
080.10.014	CyberSciTech sessions	**964 2803 9510**	**713477**
080.10.016	CBDCom sessions except CBD-08, the tutorials and CST-22	**879 1184 7709**	**536229**
080.01.002	Opening, keynotes, Panels I, R and Q, BoF and Closing, with Zoom for the keynotes and panels	**961 9676 7132**	**960648**

Keynotes



Tuesday 10 November · 09:40–10:20 · 080.01.002 · **PICom Keynote**

Machine reasoning under epistemic uncertainty: Bayesian inference and graphical models using imprecise probabilities

Dr Branko Ristic · RMIT University

Abstract Machines are increasingly asked to reason over large volumes of uncertain, heterogeneous information: sensor measurements, natural-language reports such as “possible warship,” and expert domain knowledge. Much of this uncertainty is epistemic, arising from lack of knowledge rather than randomness. Standard Bayesian inference handles it by forcing imprecise knowledge into precise probabilities, and as a result it can be confidently wrong. This talk argues for integrity in machine reasoning: a model should represent exactly the knowledge available, no more and no less. We introduce imprecise probabilities, where belief is represented by credal sets (closed convex sets of probability distributions). This framework includes belief functions, possibility theory, and random sets as special cases. We show how Walley’s generalised Bayes rule can be computed in practice using bisection and bilinear programming. We then lift it to large models through credal valuation networks, a valuation algebra that supports local computation. Examples range from classification with an imprecise confusion matrix to ongoing threat assessment onboard an underwater vehicle. In these examples, the resulting probability bounds always contain the ground truth and are tighter than those of evidential networks. We close with the computational cost and open problems.

Biography Branko Ristic received the Ph.D. degree from Queensland University of Technology in 1995. After research and engineering positions in the former Yugoslavia and in Australia, he joined the Defence Science and Technology Organisation (now Defence Science and Technology Group, DSTG) in 1996. In 2003–04 he spent a year of study leave at IRIDIA, Université libre de Bruxelles, Belgium. Since 2015 he has been a Principal Research Fellow in the School of Engineering at RMIT University, where he has led some 20 research projects, mainly funded by DSTG. He is co-author of two books, Beyond the Kalman Filter: Particle Filters for Tracking Applications (Artech House, 2004) and Particle Filters for Random Set Models (Springer, 2013), as well as over 120 journal articles and 10 book chapters. His research interests include information fusion, Bayesian estimation, target tracking, Monte Carlo methods, search techniques, sensor control, and machine reasoning under uncertainty.



Wednesday 11 November · 09:00–09:40 · 080.01.002 · **DASC Keynote**

Excursions into Distributed and Modular Quantum Computing

Prof. Seng Loke · Deakin University

Abstract The talk will introduce the idea of Distributed and Modular Quantum Computing, discuss some current trends in quantum computing, and provide an overview of a range of topics for research in Distributed and Modular Quantum Computing, as well as Quantum Internet Computing. In particular, the talk will discuss the potential of distributed and modular quantum computing to help overcome scalability challenges faced by current quantum computing architectures. It will briefly review advances in quantum computation, and the interconnection of quantum processors. The talk will then examine a range of emerging research directions in Distributed and Modular Quantum Computing, including distributed quantum operations, entanglement-assisted computation, quantum communication protocols, and techniques for coordinating computational tasks across multiple quantum processing units. The talk will also discuss the notion of the Quantum Internet and introduce Quantum Internet Computing, highlighting its relationship with Distributed Quantum Computing and its potential to enable new computational capabilities through networked quantum resources.

Biography Seng Loke is Professor in Computer Science at the School of Information Technology, Deakin University, Australia. He currently coordinates the Deakin University Quantum Computing Research Network (DQCRN), a Deakin University wide network focusing on quantum computing and its applications. Previously, he also served as an Adjunct Professor in both the School of Engineering and Mathematical Sciences at La Trobe University, and the School of Business and Tourism at Southern Cross University. Seng Loke's research interests revolve around the concept of the Internet-of-Things, ranging from distributed systems, smart/AI things and smarter cities to cooperative vehicles, edge computing, pervasive computing, IoT-assisted Agetech, ethics for smart devices, distributed quantum computing, and quantum Internet computing. He has more than 350 research publications, including books: “The Automated City: Internet of Things and Ubiquitous Artificial Intelligence” (Springer, 2021) co-authored with A. Rakotonirainy, and “From Distributed Quantum Computing to Quantum Internet Computing: An Introduction” (Wiley, Nov 2023).



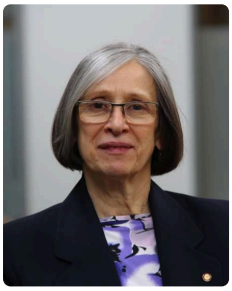
Wednesday 11 November · 09:40–10:20 · 080.01.002 · **DASC Keynote**

Machine Unlearning: Ideas and Strategies

Prof. Wanlei Zhou · City University of Macau

Abstract Over the past decade, machine learning has experienced remarkable advancements, positively impacting our daily lives. Alongside this progress, a distinctive requirement has emerged—stemming from the AI era, in particular, from privacy concerns, usability considerations, and "the right to be forgotten"—leading to the development of a new concept known as machine unlearning. In this talk, I will first give a comprehensive exploration of machine unlearning and the reasons behind the existence of this necessity. Then, I will describe the methods and technologies used in machine unlearning. Third, I will talk about developments in this field, especially through a number of our recent work on machine unlearning as examples.

Biography Professor Wanlei Zhou is currently the Vice Rector (Academic Affairs), City University of Macau, Macao SAR, China. He received the B.Eng and M.Eng degrees from Harbin Institute of Technology, Harbin, China in 1982 and 1984, respectively, and the PhD degree from The Australian National University, Canberra, Australia, in 1991, all in Computer Science and Engineering. He also received a DSc degree (a higher Doctorate degree) from Deakin University in 2002. Before joining City University of Macau, Professor Zhou held various positions including the Head of School of Computer Science in University of Technology Sydney, Australia, the Alfred Deakin Professor, Chair of Information Technology, Associate Dean, and Head of School of Information Technology in Deakin University, Australia. Professor Zhou also served as a lecturer in University of Electronic Science and Technology of China, a system programmer in HP at Massachusetts, USA; a lecturer in Monash University, Melbourne, Australia; and a lecturer in National University of Singapore, Singapore. His main research interests include security, privacy, and distributed computing. Professor Zhou has published more than 500 papers in refereed international journals and refereed international conferences proceedings, including many articles in IEEE transactions and journals. Prof Zhou is an IEEE Life Fellow.



Thursday 12 November · 09:00–09:40 · 080.01.002 · **CyberSciTech Keynote** online

Data Mining and Machine Learning for Analysis of Network Traffic

Prof. Ljiljana Trajkovic · Simon Fraser University

Abstract Collection and analysis of data from deployed networks is essential for understanding communication networks. Hence, data mining and statistical analysis of network data have been employed to determine traffic loads, analyze patterns of users' behavior, predict future network traffic, and detect traffic anomalies. The Internet has historically been prone to failures and attacks that significantly degrade its performance, affect the Internet connectivity, and cause routing disconnections. Frequent cases of various cyber threats have been encountered over the years and, hence, detection of anomalous behavior is a topic of great interest in cybersecurity. In described case studies, traffic traces collected by various collection sites are used to classify network anomalies. Various anomaly and intrusion detection approaches based on machine learning have been employed to analyze collected data. Deep learning, broad learning, gradient boosted decision trees, and reservoir computing algorithms were used to develop models based on collected datasets that contain Internet worms, viruses, power outages, ransomware events, router misconfigurations, Internet Protocol hijacks, and infrastructure failures in times of conflict. The reported results indicate that while performance of machine learning models greatly depends on the used datasets, they are viable tools for detecting the Internet anomalies.

Biography Ljiljana Trajkovic received the Dipl. Ing. degree from University of Pristina, Yugoslavia, the M.Sc. degrees in electrical engineering and computer engineering from Syracuse University, Syracuse, NY, and the Ph.D. degree in electrical engineering from University of California at Los Angeles. She is currently a professor in the School of Engineering Science, Simon Fraser University, Burnaby, British Columbia, Canada. Her research interests include communication networks and dynamical systems. Dr. Trajkovic served as IEEE Division X Delegate/Director, President of the IEEE Systems, Man, and Cybernetics Society, and President of the IEEE Circuits and Systems Society. She serves as Editor-in-Chief of the IEEE Transactions on Human-Machine Systems. She is a Distinguished Lecturer of the IEEE Systems, Man, and Cybernetics Society and was a Distinguished Lecturer of the IEEE Circuits and System Society. She is a Fellow of the IEEE.



Thursday 12 November · 09:40–10:20 · 080.01.002 · **CBDCom Keynote**

Frontiers in Specialised AI: From Vision-Language Data Pipelines to DRL-Based Compiler Optimisation

Dr Sahraoui Dhelim · Dublin City University

Abstract As artificial intelligence shifts from broad foundation models toward specialized engineering solutions, custom AI architectures are reshaping how we address domain-specific computational challenges. This keynote explores the power and adaptability of specialized machine learning paradigms across two distinct operational frontiers: automated multimodal dataset curation and low-level system optimization.

First, we examine the deployment of Vision-Language Models (VLMs) to unlock unstructured historical artifacts by automating the digitization, transcription, and semantic tagging of complex Arabic manuscripts. We demonstrate how VLMs handle intricate calligraphic styles, diacritics, and physical degradation to synthesize high-fidelity datasets at scale. Second, we pivot to systems engineering, illustrating how Deep Reinforcement Learning (DRL) navigates the massive search spaces inherent in compiler optimization, specifically phase ordering and instruction scheduling. DRL agents learn performance heuristics that consistently outpace traditional compiler passes.

By examining both domains through the lens of high-dimensional, irregular decision spaces, this talk presents actionable methodologies for designing resilient, domain-tailored AI pipelines across data-centric and compute-centric infrastructure.

Biography Dr. Sahraoui Dhelim is the research director of the School of Computing at Dublin City University. Previously, he was a senior researcher at University College Dublin from 2021 to 2024 and a Visiting Researcher at Ulster University from 2020 to 2021.

He earned his Ph.D. in Computer Science and Technology from the University of Science and Technology Beijing. He also holds a Master's degree in Networking and Distributed Systems from the University of Laghouat, and a Bachelor's degree in Computer Science from the University of Djelfa.

His scholarly contributions include over 110 peer-reviewed publications, complemented by editorial responsibilities at leading journals and conferences. He was a keynote speaker at the International Conference on Applied Artificial Intelligence and Emerging Technologies (AAIET 2025). He received the IEEE HITC Award for Excellence in Hyper-Intelligence (Technical Achievement)

His professional recognition includes Fellow HEA, he also earned the Universal Design in Teaching and Learning badge. Universal Design for Learning (UDL).

Panels

Tuesday 10 November · 10:50–12:20 · 080.01.002

Panel I · Innovation in the Age of LLMs

How will LLMs transform the way innovations are created?

Innovation has traditionally depended on human creativity, experience, and methodologies such as TRIZ. Panel I explores how LLMs, combined with data and computational intelligence, may make innovation more dynamic, scalable, and collaborative — while opening new possibilities for human-AI co-creation.

Wednesday 11 November · 10:50–12:20 · 080.01.002

Panel R · Research in the Age of LLMs

How will LLMs change the way knowledge is discovered?

From data-intensive science and the “Fourth Paradigm” to AI-assisted reasoning and autonomous scientific discovery, Panel R explores how LLMs may transform the research process itself: asking questions, forming hypotheses, connecting knowledge across disciplines, designing experiments, and accelerating discovery.

Thursday 12 November · 10:50–12:20 · 080.01.002

Panel Q · HAI-Q

Thursday 12 November · 15:30–17:00 · 080.10.013

Panel E · Education in the Age of LLMs

How will LLMs influence teaching and learning?

When knowledge is available on demand and AI can explain, question, and generate, education faces a fundamental challenge: what should humans learn, and how should they learn it? Panel E explores LLMs not merely as answer machines, but as potential tutors, Socratic partners, and tools for personalized learning and cognitive empowerment.

Monday 9 November

Online day. Online paper sessions, several running through the breaks, and three tutorials

08:00 **Registration opens** · Level 10

Parallel sessions · 09:00–10:30

DASC-01 Regular Online Session (UK, Ireland and US West Coast): Tracks 1, 2 & 4

09:00–10:20 · 080.09.012 L9 Online · Chair **Muhammad Ibrahim (QA Ltd / QA Higher Education, UK)**

09:00–09:20	**Cross-Layer Deployment Posture Assessment for Post-Quantum Cryptographic Infrastructure** Muhammad Ibrahim	#1571333040
09:20–09:40	**Reducing False Positives in Real-Traffic SQL Injection Detection via Query-Parameter Representation Alignment** Muhammed Torkey, Jawad Manzoor	#1571342510
09:40–10:00	**Protocol-Sensitive Evaluation of Log Anomaly Detection: Component Costs and Target-Access Sensitivity on HDFS and BGL** Hang Xiao, Janet Sung, Zhaoyi Li, Gangzhen Qian, Chuhong Xu	#1571335035
10:00–10:20	**RouteGraph: A BGP-EVPN Ethernet-Segment Evidence Plane for AI Job Placement** Rohan Dilip Bhosle	#1571340163

CST-01 Regular/WiP Online Session (UK, Americas, Japan and Australia): Tracks 2 & 6 and Work-in-Progress

09:00–10:05 · 080.10.013 Online · Chair **Lifei Wang (Hosei University, Japan)**

09:00–09:15	**When 100% Accuracy is Misleading: Open-Set Rejection and Compositional Generalization in WiFi Sensing** Feihuan Zhao, Kanoksak Wattanachote	#1571346547
09:15–09:35	**Silent Invalidation from Post-Publication Revision of Advisory Version Ranges** Logan Scott	#1571342809
09:35–09:55	**Evaluating Tidy-up Functional Indicators Using Virtual Reality Games** Kaoru Mitsuhashi	#1571342678
09:55–10:05	**TamperGuard: Explainable Cross-Artifact Screening for Windows Timestamp Tampering** WiP Khalid Farhan, Zihao Xu, Yuekang Li, Jingling Xue	#1571341877

CST-02 Wksp ACE-2026 Workshop

09:00–09:45 · 080.10.014 Online · Chair **Xiaohua Feng (University of Bedfordshire, UK)**

09:00–09:15	**Digital Twin-Based Cybersecurity Framework for Real-Time Threat Prediction in Smart Transport Environments: A Technical Review** Hazem Abu Adaiq, Farhinbanu Pathan, Md Israfil Biswas, Xiaohua Feng	#1571346473
09:15–09:30	**Cloud-Native Security Framework for Open Ran Using Machine Learning and Deep Learning Techniques** Migacz L., Muneer Tabassum, Xiaohua Feng, Vitaly Schetinin	#1571346531
09:30–09:45	**Collaborative Access Control Multi-keyword Searchable Encryption for Multi-Owner and Multi-User** Dequan Xu, Feng Xu	#1571335737

TUT-1a Tutorial Tutorial 1, part 1 of 2 · Resilient Delivery Pipelines, a hands-on Kubernetes tutorial

09:00–10:30 · 080.10.016 · Dr Tony de Souza-Daw, La Trobe University · Bring a laptop with VirtualBox

10:30–11:00 *Morning Tea · Level 10 (online sessions run through)*

PICOM-01 Regular/WiP Online Session (Americas): Tracks 1 to 3 and Work-in-Progress

10:30–12:20 · 080.09.012 L9 Online · Chair **Bo Wu (Tokyo University of Technology, Japan)**

10:30–10:50	**Machine Learning-Based Architecture for Intelligent Detection and Automated Response to Attacks in 5G Networks** Cleitianne Oliveira, Jose Wellington Pereira Junior, Rossana Maria de Castro Andrade	#1571333015
10:50–11:10	**Failure Prediction in Optical Sensor Time Series with Deep Learning Models** Manoel Lopes Filho, Alex Ramos, Danilo R Vasconcelos, Rossana Maria de Castro Andrade	#1571335304
11:10–11:20	**Identity Leakage Across Smartphone and Smartwatch Motion Sensors: A Privacy-Utility Study for Human Activity Recognition** WiP Paulo Guilherme de Almeida Silva, Denise Rísia Andrade Carvalho, Rossana Maria de Castro Andrade	#1571343607
11:20–11:40	**SkinVista: Examining an Augmented Reality and Machine Learning Mobile Application for Suggestive Dermatological Diagnostics in Black African** Olamiposi Olaiya, Gerry Chan, Rotimi Aderibigbe, Omololu Enigbokan, Rita Orji	#1571335428
11:40–12:00	**Enhancing Visual Preference Detection with Eye Movement Gaze Cascade and Multilayer Combinatorial Fusion** Yuanhong Wu, Yiwei Sun, Sarah Zahaykevich, Liya Fukushige, Misbah Uddin, Christina Schweikert, Frank Hsu	#1571335449
12:00–12:20	**Early-Warning Anomaly Detection Benchmarking for Cascading Failures in Cloud-Native Microservices** Vigneshwarr Venkatesan	#1571338748

CST-03 Regular Track 2: Cyber Security, Privacy & Trust (Online I)

10:30–12:30 · 080.10.013 Online · Chair **Lifei Wang (Hosei University, Japan)**

10:30–10:50	**ProTG: Multi-Granularity APT Detection and Evidence-Grounded Provenance Reasoning** Jiahao Wang, Peng Gao, Haonan Guo, Wenhao Zhang, Yibo Chen, Zhengning Pang, Chenye Zhou, Xin Jiang	#1571321671
10:50–11:10	**OCVM: Verifying Security-Constraint Repairs for AOSP Framework Patch Identification** Yibo Chen, Peng Gao, Jiahao Wang, Wenhao Zhang, Haonan Guo, Tu Mingming	#1571321945
11:10–11:30	**VaPT: Value-Guided and Feedback-Aware Planning for Automated Web Pentesting** Wenhao Zhang, Peng Gao, Kun Hu, Jiahao Wang, Yibo Chen, Haonan Guo	#1571321964
11:30–11:50	**KnowledgeAug: Semantics-Constrained Attack Subgraph Augmentation for Provenance-Based APT Detection** Haonan Guo, Peng Gao, Yuxing Wu, Wenhao Zhang, Yibo Chen, Jiahao Wang	#1571322060
11:50–12:10	**ST-XGAT: An Explainable Spatio-Temporal Graph Attention Framework for Dynamic APT Detection** Sirui Yao, Peng Gao, Qingyang Tong, Yuxing Wu, Xiang Gao, Jie Xu, Haiwei Huang, Shengzhou Wu	#1571332917
12:10–12:30	**WFA-Net A Lightweight AlexNet-Inspired CNN with Weighted Focal Loss for Multi-Class Network Intrusion Detection** Zhijia Ke, Peng Gao, Jiahao Wang, Qingyang Tong, Yu Shen, Weilong Yan	#1571334259

CST-04 Regular Tracks 4 & 5: Cyber Social Computing & Cyber Intelligence (Online I)

10:30–12:30 · 080.10.014 Online · Chair **Binbin Zhou (Hangzhou City University, China)**

10:30–10:50	**Reliability-Gated Residual Weighting for Heterogeneous Multi-Teacher Knowledge Distillation** Sifeng Wang, Kexuan Du, Chao Yan, Wanli Huang, Biao Qu	#1571334842
10:50–11:10	**Knowledge Distillation for LLM-Based Cybersecurity: Methods, Risks, and Evaluation** Kexuan Du, Sifeng Wang, Zhen Wang	#1571334910
11:10–11:30	**Budget-Aware Failure Recovery in Dynamic Code-Generation Workflows** Yan Yao, Mingzeng Zhao, Jianguo Liang, Jing Xiong, Wei Guan, Shiyou Qian, Jian Cao	#1571334979
11:30–11:50	**ASTD: Aspect-Aware Steiner-DPP for Diversified Tool Composition Recommendation from Natural-Language Queries** Hao Zhang, Xiuwen Yu, Zhaohui Cui, Sifeng Wang, Wanli Huang, Chao Yan	#1571342614
11:50–12:10	**AdaDispRec: Adaptive Diffusion Training with Label-Aware Representation Dispersion for Sequential Recommendation** Xiuwen Yu, Hao Zhang, Zhaohui Cui, Sifeng Wang, Wanli Huang, Chao Yan	#1571342696
12:10–12:30	**PERD: Personalized Exposure-aware Residual Diffusion Recommendation for Long-tail Sequential Recommendation** Xiuwen Yu, Hao Zhang, Zhaohui Cui, Sifeng Wang, Wanli Huang, Chao Yan	#1571342723

TUT-1b Tutorial Tutorial 1, part 2 of 2 · Resilient Delivery Pipelines, a hands-on Kubernetes tutorial

11:00–12:30 · 080.10.016 · Dr Tony de Souza-Daw, La Trobe University

12:30–13:30 Lunch · Level 10 (CST-05 and CST-06 start at 13:00)

Parallel sessions · 13:00–15:00

DASC-02 Regular Online Session (Asia and Australia): Tracks 1, 2 & 4

13:30–14:30 · 080.10.012 Online · Chair **Leo Yu Zhang (Griffith University, Australia)**

13:30–13:50	**From Prompt to Code: A Dual-Stage LLM-Based Security Assessment System for AI-Assisted Code Generation** Thi-Hong-Hiep Nguyen, Harikrishna Warrier, Yogesh Gupta	#1571317660
13:50–14:10	**ChangeGuard: Authorizing Identity-Baseline Updates under Organizational and Adversarial Drift** Shantanu Parashar, Shatrunjay Rawat	#1571342943
14:10–14:30	**Grouped Risk Margins in Physical Units for Dependable VPP Dispatch Screening on Unseen Feeders** Chi Cui, Jiaxing Hu, Zhijuan Jia, Haiping Du, Jia Jia	#1571342272

CST-05 Regular Track 2: Cyber Security, Privacy & Trust (Online II)

13:00–14:40 · 080.10.013 Online · Chair **Xuan Huang (Waseda University, Japan)**

13:00–13:20	**Hiding Bias in Multimodal Explanations: A Fairwashing Attack on Post-hoc Attribution** Tian Qiu, Yan Zhang, Qianmu Li, Haifeng Zhao, Yan Wang	#1571322031
13:20–13:40	**A Recursive-Decoding Enhanced CNN-BiLSTM with Self-Attention Pooling Model for Web intrusion detection** Yan Wang, Qianmu Li, Shuo Qiu, Tian Qiu	#1571324041
13:40–14:00	**A Commitment-Bound Sequential Accuracy Verification Scheme for ML Models** Chaoyue Tan, Yuling Chen, Yixian Yang, Ma Jian	#1571341397
14:00–14:20	**Cross-Lingual Generative Text Steganography via Deterministic Probability Reconstruction and Probability-Balanced Keyed Mapping** Zhang Wentao, Ma Jian, Jiayue Guan, Li Haifeng, Ren Wenshuo, Su Tao, Yuling Chen	#1571342493
14:20–14:40	**Global Attack Success Rate Understates Client-Tail Backdoor Exposure in Personalized Federated Learning** Yechan Wang, Huang Li, Kai Wang, Yuling Chen	#1571342532

CST-06 Regular Tracks 1 & 5: Cyberspace Theory & Cyber Intelligence (Online II)

13:00–15:00 · 080.10.014 Online · Chair **Binbin Zhou (Hangzhou City University, China)**

13:00–13:20	**DAWU: An Density-Aware Adaptive Windowing Updating Method for Real-Time Streaming Tensor Completion** Cong Peng, Linshu Chen, Yuxiang Chen, Shenaocheng Zhang, Liang Li, Xiangwei Meng, Zulong Diao, Wei Liang	#1571350929
13:20–13:40	**Preference-Aware Proactive Caching via Asynchronous Federated Learning for Vehicular Edge Networks** Jing Shang, Shunmei Meng, Qianmu Li	#1571313794
13:40–14:00	**ESNet: Feature-Selective Fusion Network for Building Segmentation in Remote Sensing Images** Yuexin Zhao, Guo Yuxiang, Haiyang Lin, Qi Liu, Yonghong Zhang, Xiaodong Liu, Junfei Wang	#1571323040
14:00–14:20	**DWST-NET: Dynamic Weighted Spatiotemporal Network for Long-term Radar Echo Extrapolation with Multi-source Data Fusion** Haiyang Lin, Jin Wang, Qi Liu, Yonghong Zhang, Xiaodong Liu, Junfei Wang	#1571323070
14:20–14:40	**PA-SANF: Price-Anchored Sparsity-Aware News Fusion for Robust Crude Oil Price Forecasting** Zihao Zhang, Jiaqi Liang, Ruimin Wang, Juan Xu	#1571341828
14:40–15:00	**TSRecmaNet: Temporal-Spectral Representation Refinement and Cross-Modal Alignment Network for Sales Forecasting** Jiaqi Liang, Zihao Zhang, Ruimin Wang, Juan Xu	#1571350913

TUT-2 Tutorial Tutorial 2 · Full-Duplex Spoken Dialogue for Embodied AI: Applications in Virtual Avatars and Humanoid Robot Simulation

13:30–15:00 · 080.10.016 Online · Dr Ao Guo, Nagoya University

15:00–15:30 *Afternoon Tea · Level 10 (online sessions run through)*

Parallel sessions · 15:00–17:00

PICOM-02 Regular/WiP Online Session (Korea, Europe and Israel): Tracks 1 & 4 and Work-in-Progress

15:30–17:00 · 080.10.012 Online · Chair **Antonio Guerrieri (ICAR-CNR, Italy)**

15:30–15:50	**Autonomous HVAC Control in a Real-World Building with a Multi-Agent LLM Framework** Minji An, Geonwoo Ji, Hyunkyeong Hwang, Dongwoo Kwon	#1571335050
15:50–16:10	**Toward Smart Perioperative Healthcare: A Multi-View Feature Model for Early Intraoperative Hypotension Prediction** Hesna Fartas, Ouarda Zedadra, Amina Zedadra, Mohamed Nemissi, Hadil Bencheikh, Soundous Bouhmame, Antonio Guerrieri	#1571342831
16:10–16:20	**Resilient Edge AI Architecture with Redundant Validation for Cross-Border Traffic Monitoring** WiP Przemysław Zielony, Krzysztof Piotrowski, Janusz Kaczmarek	#1571333632
16:20–16:30	**Participation Is Not Selection: A Measurement Testbed for Federated Learning over IEEE 802.15.4 Sub-GHz Links** WiP Krzysztof Turchan, Krzysztof Piotrowski	#1571342092
16:30–16:50	**Trust me, I'm a robot: Responses to uniform or differential behavior of a security robot** Michal Ezra, Ohad Inbar, Joachim Meyer	#1571342305
16:50–17:00	**Towards LLM-Enabled Cross-Domain Data Processing: A Semantic Middleware Architecture and Evaluation Framework** WiP Krzysztof Piotrowski, Igor Koropiecki, Przemysław Zielony, Krzysztof Turchan, Wiktor Zawada	#1571342647

CST-07 Regular Tracks 2 & 5: Cyber Security & Cyber Intelligence (Online III)

15:00–17:00 · 080.10.013 Online · Chair **Xuan Huang (Waseda University, Japan)**

15:00–15:20	**Cryptographic Proof of Confidentiality for Neural Network Execution on Commodity Desktop Systems as an Alternative to Conventional Cloud AI Infrastructure** Jhaswin Arjun Suresh	#1571334206
15:20–15:40	**An AI-Driven End-to-End Secure Framework for Smishing Protection in 6G Networks** Mosiur Rahaman, Nicko Cajes, Akshat Gaurav, Shavi Bansal, Kwok Tai Chui, Brij B. Gupta	#1571334548
15:40–16:00	**Ethical Boundaries and Responsibility Attribution in Cybersecurity Decision-Making Driven by Large Language Models: A CiteSpace Knowledge Mapping Analysis** Zhong Xijin, Qi Cheng Tang, Li Shu	#1571334858
16:00–16:20	**APT Detection Scheme Based on a Dual-Path Temporal-Feature Transformer** Mahmoud Basi, Tao Shang, Mustafa Mhamed, Weiwei Wang	#1571345161
16:20–16:40	**Getting the First Read Right: Breaking the Context Wall for Structured LLM Analysis of Scientific Papers** Haoyu Chen, Wenmin Lin, Xiaotong Wu	#1571351712
16:40–17:00	**Lightweight Privacy Preserving Data Aggregation for Sequentially Connected CryptoBioChip Devices** Xia Dong, Chenxi Lyu, Yaqing She, Hui Lin, Kaiwen Chen, Chen Dong	#1571340275

CST-08 Regular Tracks 2 & 6: Cyber Security & Cyber Life (Online)

15:00–16:40 · 080.10.014 Online · Chair **Brahim Walid (Hosei University, Japan)**

15:00–15:20	**Cross-Ledger Auditable Model Sharing for Federated Medical Image Learning** Yan Yao, Huihui Zhang, Biwei Yan, Junhua Wu, Guangshun Li	#1571342728
15:20–15:40	**Knowledge Distillation for Edge-Intelligent Cyber-Physical System Security: A Structured Mapping Review** Kexuan Du, Sifeng Wang, Zhen Wang	#1571350975
15:40–16:00	**EyeCoOp: Knowledge Guided Prompt Learning for Fundus Disease Classification** Jiaqi Xie, ZhiWen Wang, Cong Li, Jianuo Han, Xiaohong Yang, Lei Liu, Jinfeng Wang	#1571337377
16:00–16:20	**Multimodal Retrieval-Augmented Generation with Tongue Diagnosis for Personalized TCM Diet Recommendation** Cailin Zeng, Wenmin Lin, Lu Sun, Xiaotong Wu	#1571351701
16:20–16:40	**LactaLink: Enabling a Digital Ecosystem for Breastmilk Donation and Recipient Care** Mia Amor C Tinam-isan, Jerame B. Beloy, Christina Barazona, Nona Krislin Gomez, January Naga	#1571341475

TUT-3 Tutorial Tutorial 3 · Large Language Models for Smarter Healthcare: Recent Advances and Challenges

15:30–17:00 · 080.10.016 · Dr Ruichen Cong, Waseda University

18:00–20:00 **Welcome Reception** · 080.07.005, Level 7 balcony

Tuesday 10 November

Opening, PICom keynote, Panel I and paper sessions

08:30	**Registration** · Level 10
09:00–09:40	**Opening Ceremony** · 080.01.002
09:40–10:20	**PICom Keynote · Dr Branko Ristic (RMIT University)** *Machine reasoning under epistemic uncertainty: Bayesian inference and graphical models using imprecise probabilities · 080.01.002*
10:20–10:50	*Morning Tea · Level 10*
10:50–12:20	**Panel I · Innovation in the Age of LLMs** · 080.01.002
12:30–13:30	*Lunch · Level 10 (the online sessions below run during lunch)*

Online sessions during lunch · 12:30–13:30

CST-09 Regular Track 3: Cyber Physical Computing & Systems (Online)

12:30–13:30 · 080.10.013 Online · Chair **Yegang Du (Tohoku University, Japan)**

12:30–12:50	**Robust Shrinking-Tube Model Predictive Control for PV-Battery Building Energy Management** Nanyan Gai, Chengdian Yang, Stefano Mazzoni, Ke Yan	#1571322903
12:50–13:10	**Latent Semantic Goal-Driven Navigation Framework for Indoor Environments** Jianguo Zhao, Xinqi Yang, Qiao Pan, Mei Wang	#1571334582
13:10–13:30	**Robust and Efficient Multi-Paxos for Blockchain-enabled Data Synchronization in Cloud-Edge Collaboration** Wenmin Lin, Zihao Chen, Chenxi Liu, Xiaotong Wu	#1571351703

CBD-01 Regular Online Session: Track 2 (I)

12:30–13:30 · 080.10.016 Online · Chair **Haoxiang Luo (Nanyang Technological University, Singapore)**

12:30–12:50	**HypeGuard-FL: A Dual-Branch Structural Sanitization Framework for Secure Federated Hypergraph Learning** Zhuoyue Li, Xinjun Pei, Qimeng Yang, Weishang Wu, Peiyuan Guan, Zhao Yunlong	#1571324073
12:50–13:10	**Federated Critical Parameter Synchronization with Endogenous Self-Distillation for Heterogeneous Graph Learning** Maoxu Wang, Xinjun Pei, Jinmiao Song, Ping Jiang, Weishang Wu, Zhao Yunlong	#1571324212
13:10–13:30	**Authentication and Key Management Scheme for Task-Oriented Multi-Agent Environments** Chenxu Wang, Ying Zhu, Fangfang Dang, Boyu Liu, Zhaotai Meng, Xuanang Ding, Peng Lin	#1571342552

Parallel sessions · 13:30–15:10

PICOM-03 Regular Tracks 4 & 5: Next Gen Smart Environments & Edge Intelligence Applications

13:30–15:00 · 080.10.012 Hybrid · Chair **Claudio Miceli de Farias (Federal University of Rio de Janeiro, Brazil)**

13:30–13:50	**Tupã: An Approach for Anomaly Detection in Non-IID Smart Grid Data Using Federated Learning** Juliana Fernandes Dal Piaz, Julia Rocha, Pablo Rangel, Claudio Miceli de Farias	#1571343066
13:50–14:10	**Adaptative Random Sampling Applied to an Unbalanced Scenario** Mayara Aragão, Mariana Aragão, Juliana Fernandes Dal Piaz, Claudio Miceli de Farias	#1571343068
14:10–14:30	**EVEREST: Indoor Presence Detection Using COTS Wi-Fi Devices and Passive Sensing** Francisco Laurindo C. Junior, Rossana Maria de Castro Andrade, Tales Paiva Nogueira	#1571335258
14:30–14:50	**Continual Graph Learning in the Edge-Cloud Continuum via Model Souping** online Khaing Wah Swe, Thanda Shwe, Israel Mendonca, Masayoshi Aritsugi	#1571341864

CST-10 Regular Track 1: Cyberspace Theory & Technology

13:30–15:00 · 080.10.013 Hybrid · Chair **Lai Tu (Huazhong University of Science and Technology, China)**

13:30–13:50	**Joint Traffic-State Estimation and Forecasting with a Spatiotemporal Heterogeneous Graph Neural Network** Zilang Wang, Benxiong Huang, Lai Tu	#1571324631
13:50–14:10	**D-SFC: Workload-Aware Dual-Reference Search for 2D Piecewise Space-Filling Curves** online Yan Yao, Shiyou Qian, Jian Cao, Jiaojiao Li, Zhaotang Hou, Hao Zhang, Wenshun Wang	#1571331901
14:10–14:30	**3D Scene Reconstruction and VR Space Generation Using Panoramic Depth Data Captured by a Smartphone** Hiroki Kobayashi, Katashi Nagao	#1571332638
14:30–14:50	**Multi-Criteria Decision Optimization for Slice-Aware Intermediate UPF Placement** Yaseen Al Doori, Leo Popokh, Raul Rodriguez, Suku Nair	#1571342785

CST-11 Regular Tracks 2, 5 & 6: Cyber Security, Cyber Intelligence & Cyber Life

13:30–15:10 · 080.10.014 Hybrid · Chair **Ali Dewan (Athabasca University, Canada)**

13:30–13:50	**MedFlowAgent: An LLM-Based Agent for Intelligent Medication Workflow Management to Assist Ward Nurses** online Xiaoyan Zhang, Chenyang Li, Ruichen Cong, Zhihe Zhang, Zhao Rui, Qingguo Zhou, Qun Jin	#1571334014
13:50–14:10	**Accident Severity Prediction Using Federated Learning and Synthetic Data Augmentation** online David Jackman, Dunwei Wen, M. Ali Akber Dewan	#1571335130
14:10–14:30	**CSAM: Segment Anything Model Meets Concept Learning for Scene Parsing of Agricultural Landscapes** Zhongqi Lin, Zengwei Zheng, Shenfei Pei, Yunhan Zheng, Bo Sun, Si Yehuan	#1571341794
14:30–14:50	**A Framework for Making Physical Objects into AR- and LLM-based Interactive Physical Agents** Kota Sakai, Akishige Yuguchi, Yoshio Matsumoto	#1571324144
14:50–15:10	**Beyond Random Splits: A Leakage-Resistant Evaluation Protocol for Cross-Source Deceptive Email Detection** Victor Chang, Weihang Huang, Hai Wang, Liucheng Guo	#1571341554

CBD-02 Regular Track 4: Networking and Communications

13:30–15:00 · 080.10.016 Hybrid · Chair **Yibin Zhou (Shanghai Jiao Tong University, China)**

13:30–13:50	**MGDFormer: A Modality-Gated Distributional Transformer For Electricity Price Forecasting** online Xiaoxin Huang, Meize Chen, Chenhao Wu, Zheng Wen, Hiroshi Onoda, Jiang Liu	#1571323794
13:50–14:10	**Adaptive Access Point Scheduling for Seamless Communication in Dense and Dynamic Edge Environments** Feng Hanxuan, Kazunori Sugiura	#1571331907
14:10–14:30	**ReDirect: LLM-Guided Directional Neural Resets for Multi-Agent UAV Emergency Communications** Wen Qiu, Xun Shao, Wai Shiang Cheah, Wei Zhao, Hiroshi Masui	#1571332166
14:30–14:50	**Codebook-Aided JSCC for Image Transmission Toward MLLM-Based Understanding in IoV** Shilong Zhang, Jinhua Chen, Tao Yu, Tong Liu, Weiyu Wang, Franck Junior Aboya Messou, Keping Yu	#1571360765

15:00–15:30 *Afternoon Tea · Level 10 (CST-11 ends at 15:10)*

DASC-03 Regular Track 2: Security, Privacy, and Trust in Systems and Networks

15:40–17:00 · 080.10.012 Hybrid · Chair **Wanlun Ma (Swinburne University of Technology, Australia)**

15:40–16:00	**A TEE-Protected Framework for Explainable LLM-Assisted Microgrid Decision Support** Fusen Guo, Zhibo Zhang, Wei Song, Haoxiang Tian, Jun Zhang, Mo Huadong	#1571323299
16:00–16:20	**Unsupervised Anomaly Detection in Industrial Internet of Things: What's Next?** Mariya Halitim, Talal Halabi, Martine Bellaïche, Steven Ding	#1571332757
16:20–16:40	**Automated Zero-Day Vulnerability Detection and Patching via Prompt-Optimized Language Models** Aziz Srairi, Oussama El Boujjadia, Fehmi Jaafar, Haïfa Nakouri, Manel Abdelkader	#1571335266
16:40–17:00	**Policy-Bound Encrypted Objects: Dependable Metadata Binding and Rollback-Aware Storage** online Mario Raso, Daniele Friolo	#1571335550

CST-12 Regular Tracks 2 & 3: Cyber Security & Cyber Physical Computing

15:30–17:05 · 080.10.013 · Chair **Siyang Lu (Beijing Jiaotong University, China)**

15:30–15:50	**FAWM: Frequency-Domain Watermarking Backdoor Attack against Conditional Generative Models** Yuetong Lu, Kai Zhao, Yuke Xiong, Siyang Lu, Wei Lu	#1571342699
15:50–16:10	**PUMA: Dual-Uncertainty Regularization for Defending Against Model Inversion Attacks** Dehua He, Kang Han, Di Wu, Gaowen Liu, Ramana Rao Kompella, Wei Xiang	#1571345051
16:10–16:30	**S2D-ODE: Sparse-to-Dense Neural ODE for Full-Field Tunnel Response Perception under Sparse Sensing** Zicong Zhao, Mingxuan Xie, Junchen Ye, Xuyan Tan, Bowen Du	#1571322381
16:30–16:50	**Weight-Aware Resequencing of Electric Truck Platoons for Battery Energy Balancing** Boyu Shu, Zhiwu Huang, Feng Zhou, Tianji Lv, Hao Lin, Cunrui Wang, Weirong Liu	#1571332213
16:50–17:05	**Towards DisasterFM: Foundation Models for Disaster Science** Yilin Lyu, Zhong Chen, Kang Liu	#1571342446

CST-13 Regular Tracks 1, 4 & 6: Cyberspace Theory, Cyber Social Computing & Cyber Life

15:30–17:00 · 080.10.014 Hybrid · Chair **Yishui Zhu (Chang'an University, China)**

Talk 1	**Joint Orchestration and Serverless Scheduling for Dynamic Agentic Workflows** online Yan Yao, Huihui Zhang, Junhua Wu, Guangshun Li	#1571342917
Talk 2	**Persona and Few-Shot Prompting for Large Language Model-Based Cyber-Social Analysis of Tourism Motivation** Zhenzhen Liu, Fumito Masui, Juuso Eronen, Michal Ptaszynski	#1571334966
Talk 3	**An Educational Behavior Analysis Framework Based on Spatiotemporal Event-Evidence** online Jingsen Huang, Yishui Zhu, LuYang Wang, James Gwenani	#1571331338
Talk 4	**A Causal Context-Augmented Framework for Personal Health Prediction Enabled by LLMs** Ruichen Cong, Zecheng Ren, Yinghan Xu, Shoji Nishimura, Atsushi Ogihara, Qun Jin	#1571350232
Talk 5	**Spatial Resolvability-Guided Vital-Sign Separation for Two Adjacent People Using FMCW Radar** Xiangshu Qi, Jianhua Ma	#1571346614

CBD-03 Regular Tracks 1, 3 & 4: Theories, IoT and Networking

15:30–17:00 · 080.10.016 · Chair **Yibin Zhou (Shanghai Jiao Tong University, China)**

15:30–15:50	**Provenance-Gated Agent Collaboration for Multimodal IoT Intrusion Detection** Yibin Zhou, Jianhua Li	#1571335041
15:50–16:10	**Understanding and Controlling Shared-Endpoint Message Passing in Traffic Interaction Graphs** Yibin Zhou, Chaofeng Zhang, Yijun Lu, Xiao Yang, Gaolei Li, Jun Wu, Jianhua Li	#1571335390
16:10–16:30	**A Scheduling Method for Division-Based Broadcasting Considering Error Resilience** Yusuke Gotoh	#1571333681
16:30–16:50	**Same Footage, Opposite Sign: Cadence, Coverage and Cancellation in UAV Video Counting** Kou Xuri, Musen Cheng, Taihui Liu, Hiroshi Masui	#1571335555

Wednesday 11 November

Two DASC keynotes, Panel R, special sessions, workshops, paper sessions, banquet and awards

08:30	**Registration** · Level 10
09:00–09:40	**DASC Keynote · Prof. Seng Loke (Deakin University)** *Excursions into Distributed and Modular Quantum Computing · 080.01.002*
09:40–10:20	**DASC Keynote · Prof. Wanlei Zhou (City University of Macau)** *Machine Unlearning: Ideas and Strategies · 080.01.002*
10:20–10:50	*Morning Tea · Level 10*
10:50–12:20	**Panel R · Research in the Age of LLMs** · 080.01.002
12:30–13:30	*Lunch · Level 10 (the online session below runs during lunch, and DASC-07 starts at 13:00)*

Online sessions during lunch · 12:30–13:30

CBD-04 Regular Online Session: Track 1

12:30–13:30 · 080.10.016 Online · Chair **Yinqiu Liu (Nanyang Technological University, Singapore)**

12:30–12:50	**GLAE²-Nets: Global-Local Autoencoder² Networks with Orthogonal Constraints for Multi-View Clustering** Mei Lu, Xiaolu Zhao, Xiangjun Zhao	#1571332666
12:50–13:10	**ReCAST-YOLO: Recalibrated Cross-Scale Fusion with Mixed Local Channel Attention for Small Traffic Sign Detection** Han JiZhou, Zhiyu Wu, Shenghui Cui, Zhenjun Dong, Sudan Xin, Zunwang Ke	#1571340946
13:10–13:30	**Smart Cocoa Pod Monitoring System Using Multitask Deep Neural Network** Daniel Chee Ho Ong, Wai Shiang Cheah, Xun Shao, Nurfauza Jali, Navies Maisin	#1571341357

Parallel sessions · 13:00–15:00

DASC-07 Wksp EDCSTA 2026 Workshop

13:00–13:30 · 080.10.012 · Chair **Bernady O. Apduhan (Kyushu Sangyo University, Japan)**

13:00–13:15	**IoT-Enabled Self-Adaptive Edge AI for Strawberry Defect Detection with Pseudo-Labeling and Continual Learning on Jetson Orin Nano** Tong Jun Yang, Kazuaki Tanaka, Way Soong Lim, Krishnamoorthy R	#1571337307
13:15–13:30	**Evaluating Stash-Based Caching in Tree-Based ORAMs** Kazuaki Kawabata, Naohiro Hayashibara	#1571342465

DASC-04 Regular/WiP Tracks 1 & 3 and Work-in-Progress

13:30–15:00 · 080.10.012 · Chair **Yao Hu (The University of Tokyo, Japan)**

13:30–13:50	**Fault-Tolerant Lossy-Compressed Allreduce for Approximate Collective Communication** Yao Hu	#1571315246
13:50–14:10	**Evolutionary Generation of Hardware-Trojan Variants in Gate-Level Netlists via Circuit Redundancy** Sho Yoshimi, Nozomu Togawa	#1571333522
14:10–14:30	**Efficient Container Restoration outside Heavily Loaded VMs** Shota Kimoto, Kenichi Kourai	#1571335515
14:30–14:50	**Autonomic UPF Selection in 5G Core Networks Using GRASP** Bilal Ghani, Amine Fassihi, Hakima Chaouchi	#1571342755
14:50–15:00	**Per-Chunk Multipath Dispersal for CCNx Content Retrieval over Heterogeneous Paths** WiP Junichi Funasaka	#1571346607

CST-14 SS SecTrust-AI Special Session: Security and Trustworthiness of Large AI Models and Agents (I)

13:30–15:00 · 080.10.013 Hybrid · Chair **Yukuan Yang (Chinese Academy of Sciences, China)**

13:30–13:45	**Preliminary Report On A Resource-Aware Automated Framework for Indirect Prompt Injection on HTML Accessibility Trees** online Calin Anton, Dana Cobzas, Benjamin Lawrence, Nolan Schlacht	#1571346363
13:45–14:00	**Towards Lifecycle-Aware Software Composition Analysis: An Empirical Study** online Chaofan Li, Yutao Hu, Yueming Wu, Deqing Zou	#1571348300
14:00–14:15	**RoadMap+: LLM-Driven Completion of a Universal Data Poisoning Attack Knowledge** Tarek Chaalan, Shaoning Pang, Joarder Kamruzzaman	#1571346663
14:15–14:30	**SCORE: Continuous-Output Spiking via Non-Reset Membrane Potentials for Energy-Efficient EEG-Based Affect Tracking** Yang Chi, Man Yao, Chengxi Xie, Jihong Wang, Shile Qi, Yuzhe Zhang	#1571347859
14:30–14:45	**EGMCJ: Evidence-Guided Minority Correction for Multi-LLM Judge** Weiqi Mao, Wenli Zhang, Xuanzhe Liu, Tiechui Yao, Bo Chai, Mengjie Shi	#1571348421
14:45–15:00	**A Layered and Phase-Oriented Security Framework for LLM Agents** Yukuan Yang, Chen Yang	ID TBC

CST-17/19 SS/Wksp SecTrust-AI Special Session (II)/HMGI-CPSS 2026 Workshop

13:30–15:00 · 080.10.014 Online · Chair **Yegang Du (Tohoku University, Japan)**

13:30–13:45	**Enabling Usage Control for Transformers via Keyed Dense Structural Mixing** Jinyong Zhang, Lisha Yao, Zhi Li	#1571348647
13:45–14:00	**Foreground-Adaptive Memory Bank Learning for Object-Centric Industrial Anomaly Detection** Jiali Wang, Jiani Wei, Bowen You, Haokun Wang, Zhuxin Si, Mingrui Hou, Jing Liang	#1571341921
14:00–14:15	**Detection-Oriented Underwater Image Enhancement with Object-Frequency Preservation** Huaxu Li, Ting Wang, Zishuai Wang, Xuanzhen Zhang, Yuan Fang	#1571342741
14:15–14:30	**Crop Protection and Two-Dimensional Point Selection for UAV Weed Perception** Ting Wang, Huaxu Li, Zishuai Wang, Yugang Yang, Yuan Fang	#1571342742
14:30–14:45	**Illumination-Aware Temporal Consistency Refinement for Low-Light Video Enhancement** Zishuai Wang, Huaxu Li, Ting Wang, Wenbo Wu, Yuan Fang	#1571342886
14:45–15:00	**Early Failure Prediction for LLM Agent Execution Chains Using Multidimensional Runtime Features** Wenbo Wu, Aojie Xu, Yuan Fang	#1571342732

CBD-05 Regular Track 2: Systems, Management and Security

13:30–15:00 · 080.10.016 · Chair **Siyang Lu (Beijing Jiaotong University, China)**

13:30–13:50	**Multi-Cloud Provider Feature Compatibility Metrics** Dac Tuan Thanh Nguyen, Tony De Souza-Daw	#1571342039
13:50–14:10	**FedFold: Hard-Case-Aware Federated Parameter-Efficient Adaptation for Protein Structure Prediction** Bin Wu, Jue Wang, Xuanmeng Sha, Bingchang He, Weiwei Hu, Fumihiko Ino	#1571342460
14:10–14:30	**Cloud-assisted Generative Replay for Continual Learning in Edge Intelligence** Thanda Shwe, Masayoshi Aritsugi	#1571342513
14:30–14:50	**UAP-DB: Latent-Trigger Backdoor Attacks on DreamBooth Personalization** Yuke Xiong, Jingyao Xu, Siyang Lu, Wei Lu	#1571342597

15:00–15:30 *Afternoon Tea · Level 10*

Parallel sessions · 15:30–17:00

PICOM-04 Regular/WiP Tracks 2 & 3, Regular Paper and Work-in-Progress

15:30–17:00 · 080.10.012 · Chair **Claudio Miceli de Farias (Federal University of Rio de Janeiro, Brazil)**

15:30–15:50	**Blur-Aware Video Deblurring via Optical Flow Guidance** Ying Fu, Jifan Zhang, Jing Peng, Xiaojie Xiaojie Li, Jiliu Zhou	#1571319896
15:50–16:10	**AOE: Adversarial Optimized bias Elimination for embedding-based Recommender System** Haofeng Li, Jinyao Xu, Ian G. Harris	#1571321855
16:10–16:30	**GeoCausal-Alert: A Counterfactual Causal and Geospatially Consistent Early-Warning System for Hydrological Hazards** Saddaf Rubab, Hafsa Waheed	#1571342630
16:30–16:50	**Attention-Based Intermediate Feature Fusion Network for Robust Retinal Disease Classification from Multimodal OCT Data** Dhiya Sewnundan, Alfa P Eleojo, Marion O Adebiyi, Nalindren Naicker	#1571345505
16:50–17:00	**Reservoir Computing for Urban Air Quality Forecasting: A Comparative Study of Recurrent Architectures** WiP Francesco D'Amore, Delia Evelina Bruno, Antonio Guerrieri, Francesco De Simone, Carlo Mastroianni, Andrea Vinci, Franco Cicirelli, Mariantonia Bencardino	#1571345625

CST-15/16 SS LM-CSC&CEA/Cyber-IoT Special Session

15:30–17:00 · 080.10.013 Hybrid · Chair **Rui Xing (Jingdezhen Ceramic University, China)**

15:30–15:45	**An Isaac Sim-based Framework for Synthetic Scene Generation and Automatic Annotation Generation** online Qi Zhang, Muxing Li, Mei Wang, Qiao Pan	#1571334561
15:45–16:00	**LLM Harness Engineering for Regulatory Compliance: Domain-Agnostic Rules Derived from First Principles** online Shaning Pang, Muhamad Magableh, Suryani Lim	#1571344579
16:00–16:15	**Large Language Model-Based Patient Agent Generation for Cognitive Disorders** Fengwei Cai, Qiao Pan, Mei Wang, Dehua Chen	#1571341871
16:15–16:30	**Task-Priority-Aware Uplink NOMA Power Allocation for UAV Swarm-LEO Satellite Communications** online Qingning Shi, Qichao Xu	#1571342103
16:30–16:45	**GridMoE: Context-Aware Anomaly Detection in Topology-Varying Power Grids** Mingwei Xu, Xiaoliang Wang, Qinnan Hu, Linkang Du	#1571351291
16:45–17:00	**A Digital Twin-Driven Collaborative Abnormal Sensing Framework for Metaverse-Assisted Internet of Vehicles** Rui Xing, Minghui Dai, Jiange Jiang, Kehan Huang	#1571345572

CST-18 Wksp HAIDC 2026 Workshop

15:30–17:00 · 080.10.014 Hybrid · Chair **Toshimi Ogawa (Tohoku University, Japan)**

15:30–15:45	**Keeping Older Adults in Control: A Human-Centred Conversational Health Coach for Movement at Home** online Justin Gallagher	#1571341697
15:45–16:00	**From Social Potential to Deployment Requirements: Evaluating a Culturally Adapted Conversational AI Avatar in Japanese Elderly Care** Eva Theresa Jahn, Moritz Hartung, Yegang Du, Toshimi Ogawa, Rainer Wieching, Dave Randall, Yasuyuki Taki, Volker Wulf	#1571336921
16:00–16:15	**A Feasibility Study of a Personalized Social Participation Support Platform: Usability Evaluation and Requirements Identification for the Development of a Virtual Link Worker** Toshimi Ogawa, Yegang Du, Parastou Azari Gargari, Jasmin Lehmann, Rainer Wieching, Yuzuru Nakagaki, Yasuyuki Taki	#1571341344
16:15–16:30	**Design and Preliminary Evaluation of an LLM-Based Social Event Recommendation and Sharing System for Older Adults** Yegang Du, Toshimi Ogawa, Gabriele Trovato, Rainer Wieching, Yasuyuki Taki	#1571341356
16:30–16:45	**Designing an AI Companion App to Facilitate Real-World Social Engagement Among Senior Populations** Jessica Takabayash, Toshimi Ogawa, Yegang Du, Yasuyuki Taki	#1571342724

CBD-06 Regular Tracks 2 & 3 and Regular Papers

15:30–17:00 · 080.10.016 Hybrid · Chair **Siyang Lu (Beijing Jiaotong University, China)**

15:30–15:50	**Vehicle Trajectory Forecasting with Dynamic Sparse Spatio-Temporal Graph Neural Networks** online Youde Wu, Tingyan Long, Changheng Li	#1571346366
15:50–16:10	**How Much Regularization Survives Averaging? Update Masking in Federated Learning** online Wenhao Yan, Fu Kuroda, Yucheng Jin, Zhenke Chen	#1571346387
16:10–16:30	**A Low-Cost Multi-View RGB Sensing System for Markerless 3D Human Pose Reconstruction in Smart Manufacturing** online Li Quanhe, Wei Wang, Moyan Liu, Shigeki Sugano	#1571342138
16:30–16:50	**Breaking the Inner Loop: Efficient Synchronous Co-Training for Identity Protection Against Personalized Diffusion Models** Yiran Qi, JingYao Xu, Siyang Lu	#1571342641

18:30–21:30 **Conference Banquet & Awards** · venue TBA

Thursday 12 November

CyberSciTech and CBDCom keynotes, Panel Q, workshops and Panel E

08:30	**Registration** · Level 10
09:00–09:40	**CyberSciTech Keynote · Prof. Ljiljana Trajkovic (Simon Fraser University)** online *Data Mining and Machine Learning for Analysis of Network Traffic · 080.01.002*
09:40–10:20	**CBDCom Keynote · Dr Sahraoui Dhelim (Dublin City University)** *Frontiers in Specialised AI: From Vision-Language Data Pipelines to DRL-Based Compiler Optimisation · 080.01.002*
10:20–10:50	*Morning Tea · Level 10*
10:50–12:20	**Panel Q · HAI-Q** · 080.01.002
12:30–13:30	*Lunch · Level 10 (DASC-05 starts at 13:00)*

Parallel sessions · 13:00–15:00

DASC-05 Wksp ADSN 2026 Workshop

13:00–15:00 · 080.10.012 Hybrid · Chair **Junichi Funasaka (Hiroshima City University, Japan)**

13:00–13:30	**Invited Talk · Toward Reliable and Efficient Transport in Information-Centric Networking** online Dr Kazuhisa Matsuzono (NICT, Japan)	
13:30–13:45	**Low-Frequency Changes in Country-Name Search Trends Around Major International Crises: A Cross-Case Comparison and an Iran Case Study** Yu Chen, Kazuki Nakajima, Masaki Aida	#1571339170
13:45–14:00	**Cluster-Based Decentralized Federated Learning for Dynamic Wireless Network Environments** Yuki Maeda, Tomoyuki Ohta	#1571339200
14:00–14:15	**Introducing IP Multicast into CCNx Content Object Distribution over a Consumer-Side LAN** Junichi Funasaka	#1571341474
14:15–14:30	**Low-Frequency Features of Stock-Related Search Interest and Subsequent Price Movements in Japanese Automakers** Takumi Ishii, Chisa Takano, Hiroyasu Obata, Masaki Aida	#1571341799
14:30–14:45	**Autonomous Decentralized Transmission Control Based on a Coupled Oscillator Model for Federated Learning with Non-IID Data** Yuto Teshima, Chisa Takano, Hiroyasu Obata, Satoru Kobayashi, Yasuhiro Ohishi	#1571342252
14:45–15:00	**Aggregation-Based Candidate Monitoring Keyword Selection for Trend Precursor Analysis** Chihiro Sanada, Kazuki Nakajima, Masaki Aida	#1571346303

CST-20 Wksp GAI-Hyperl 2026 Symposium (I)

13:30–15:00 · 080.10.013 · Chair **Ruichen Cong (Waseda University, Japan)**

13:30–13:45	**Reshaping Research, Innovation, and Education in the Age of LLMs** Qiangfu Zhao, Qing Li, Zhong Chen, Wei Lu, Jiliu Zhou, Xiao Huang, Qun Jin, Prem Prakash Jayaraman, Seng W. Loke, Naohiro Hayashibara, Toshiaki Miyazaki, Jiannong Cao, Lizhen Cui, Claudio Miceli de Farias, Benxiong Huang, Zengwei Zheng, Zhen Liu, Chen Yang, Hiroshi Masui, Jianhua Ma	#1571345378
13:45–14:00	**Toward Human-State Inferability in LLM-Based Classification Using Multimodal Wearable Data** Lifei Wang, Ao Guo, Jianhua Ma	#1571346229
14:00–14:15	**Relevant ≠ Applicable: Diagnosing the Conditional Applicability Illusion in Retrieval-Augmented Regulation QA** Shitian Zheng, Zengwei Zheng, Zhengzhe Xiang, Binbin Zhou	#1571341777
14:15–14:30	**Position: Toward Interleaved Hardware-Model Development for AI Systems** Ruilin Zhang, Zhong Chen	#1571342464
14:30–14:45	**Beyond Sentiment: Comparing Affective Polarity and Stance in Japanese YouTube Comments on Human Papillomavirus Vaccination** Yinghan Xu, Ruichen Cong, Mai Yaguchi, Masahiko Sakaguchi, Kayoko Katayama, Qun Jin, Shoji Nishimura, Atsushi Ogihara	#1571343232

CST-21 Wksp IoT-Life 2026 Workshop (I)

13:30–15:00 · 080.10.014 · Chair **Yishui Zhu (Chang'an University, China)**

13:30–13:45	**Effects of AI Voice Prompt Volume on Drivers' Braking Reaction Time, Heart Rate Responses, and Pupil diameter Responses while Driving Simulation** Zhenzheng Xu, Bo Wu	#1571334610
13:45–14:00	**Eye Movement-Based Analysis of Children's Reasoning Performance via Chinese Character Evolution Ordering Experiments** Qun Xu, Bo Wu, Letian Liu, Qianyue Xu	#1571334612
14:00–14:15	**Effects of Driving Experience on Eye Movement Metrics Under Different Multi-Frequency Music Combinations during Prolonged Driving** Chenhao Li, Bo Wu, Jun Wang	#1571334613
14:15–14:30	**Eye-Tracking-Based Analysis of Attention Mechanisms in Video-based Learning: How Students' Mental Health Status Affects Their Learning of New Interdisciplinary Concepts** Xinyue Yang, Bo Wu, Letian Liu, Jun Wang, Siyu Xiong	#1571334623
14:30–14:45	**Effects of Pictorial Representation Styles on Visual Attention in Preschool Children** Qianyue Xu, Bo Wu, Qun Xu, Letian Liu	#1571334776
14:45–15:00	**Knowledge Graph-Guided Temporal Behavior Interpretation with A Local LLM Agent in Smart Homes** Tianle Hu, Hong Chen, Yenjou Wang	#1571336970

CBD-07 Regular Online Session: Tracks 2 to 4

13:30–14:50 · 080.10.016 Online · Chair **Zhen Tian (University of Glasgow, UK)**

13:30–13:50	**A PUF-Assisted Lightweight Access Authentication and Key Agreement Protocol for Edge-Assisted Smart Grids** Chang Liu, Jiakai Hao, Jun Yun, Xu Dong, Tongjia Zhang, Lvxuan Zhou	#1571342605
13:50–14:10	**Lightweight Indirect Prompt Injection Detection via Executor-Aware Semantic Difference** Yadong Wang, Guangtao Zhou, Jinliang Gong, Hongyang Ruan	#1571340038
14:10–14:30	**BoundarySpace: Executable Sovereignty for AI Agents in Data Spaces** Shuwen Liu, George C. Polyzos	#1571341817
14:30–14:50	**Adaptive Hierarchical PQ-GPS Scheduling for Data-Intensive Services in Mobile Networks** Ran Zhang, Huaifeng Sun, Tong Ding	#1571333793

15:00–15:30 *Afternoon Tea · Level 10 (CBD-08 starts at 15:20)*

Parallel sessions · 15:20–17:00

DASC-06 Regular Track 4 (AI, Machine Learning, and Advanced Computational Methods) and Regular Paper

15:30–16:50 · 080.10.012 · Chair **Fusen Guo (Swinburne University of Technology, Australia)**

15:30–15:50	**Machine Learning-Assisted Multiobjective Trajectory Optimization for an Industrial Robot Using a Lightweight Digital Twin** Ashan Kudabalage, Supun Dissanayaka	#1571332391
15:50–16:10	**A Graded Mechanism Atlas for Continuous Casting: Auditable Evidence for LLM Causal Diagnosis** Jike Li, Haiping Du, Yujun Zhang, Yuexiao Guan, Yan Yu	#1571333576
16:10–16:30	**A Robust Watermark-based Fingerprint Framework for GNNs Ownership Verification** Han Zhang, Yan Wang, Liu Guanfeng, Pengfei Ding, Huaxiong Wang, Kwok-Yan Lam	#1571342945
16:30–16:50	**Feature-Space Flatness and Adversarial Transferability** Norihiro Shimada, Seiichi Ozawa, Hiroaki Inoue, Tao Ban, Geon-Kyu Bang, Takeshi Takahashi	#1571346313

Panel E Panel Education in the Age of LLMs

15:30–17:00 · 080.10.013 Hybrid

CST-23 Wksp/WiP IoT-Life 2026 Workshop (II), with Work-in-Progress talks

15:30–17:00 · 080.10.014 Hybrid · Chair **Bo Wu (Tokyo University of Technology, Japan)**

15:30–15:45	**Instance-Bound Regional Evidence Watermarking for Cross-Platform Consistency Verification of Personal IoT Images** online Chongyang Zhang, Yu Qi, Bo Wu	#1571337618
15:45–16:00	**Multi-Scale Transformer Autoencoder with Haar-based Local Frequency Enhancement and Pseudo-Defect Guided Denoising for Industrial Anomaly Detection** online Hao Li, Qien Yu, Yan Tan, Bo Wu	#1571342195
16:00–16:15	**A Cross-Modal Knowledge Distillation Framework for Smart-Insole-Based Daily Walking Task State Prediction** Xuan Huang, Yanmei Jiao, Yuan Bian, Xinyue Yang, Shoji Nishimura, Bo Wu	#1571345680
16:15–16:30	**Analyzing Temporal Changes in Drivers' Pupil Diameter Around Tunnel Entry** Zijun Wang, Bo Wu, Yishui Zhu, Qirun Wang, Qun Jin	#1571346650
16:30–16:40	**Multi-UAV Cooperative Yak Search Based on MAPPO for Intelligent Pasture IoT** WiP online Qilin Zhang, Caixing Shao, Ran Li, Hang Lin, Jianhua He	#1571334268
16:40–16:50	**How do Immersive Mirror-Reversal Views and Embodiment Affect the Ease of Teleoperated Bring-me Tasks using a Mobile Manipulator?** WiP Keiichi Ono, Akishige Yuguchi, Yoshio Matsumoto	#1571342957
16:50–17:00	**A Methodological Framework for Developing and Evaluating Visual Representations of Mental Imagery** WiP Takuya Ohmori	#1571346252

CST-22 Wksp GAI-Hyperl 2026 Symposium (II)

15:30–17:00 · 080.10.016 Hybrid · Chair **Yinghan Xu (Waseda University, Japan)**

15:30–15:45	**Extracting Short-Term Intention Contrastive Learning for Sequential Recommendation** online Mengfan Ma, Zhenhai Wang, Qingxing Wu, Xian Wu, Yuhao Xu	#1571324399
15:45–16:00	**A Human-AI Collaborative, Verifiable Corpus Analysis Paradigm and Its Instantiation on a Sino-Japanese Conceptual-History Study** online Haozhong Duan, Shuliang Zhu, Ming Wang, Raymond Yao, Xiaolin Wu	#1571342189
16:00–16:15	**Automated Scientific Ideation: A Review and Future Perspectives** online Zhipeng Luo	#1571342852
16:15–16:30	**NeuroCoSci: Human-AI Hyper-Intelligence for Mechanism-Grounded Multimodal Neuroimaging** Xing-chen Zhou, Shan-Yan Lei, Zhao Yao, Zuo-Bing Chen	#1571329782
16:30–16:45	**Beyond Human Inspection: TRACE for Provenance-Aware Evaluation of AI-Assisted Text Pipelines** Antonis Charalampous, Constantinos Djouvas, Nicolas Tsapatsoulis, Christos Christodoulou	#1571339485
16:45–17:00	**Memory-Enhanced Vision-Language Alignment for Image Aesthetic Assessment** Xixi Nie, Shixin Huang, Xiaoyu Wan	#1571341291

CBD-08 Regular Online Session (Asia, Middle East and Europe): Tracks 1, 2 & 4

15:20–17:00 · 080.09.012 L9 Online · Chair **Zhen Tian (University of Glasgow, UK)**

15:20–15:40	**Quantum Space-Time Block Codes for Noncoherent Communication: A Simulation-Based Comparison with Stabilizer Codes** Mayasa Khalifa Al Hinai, Hafiz Asif, Abdulnasir Hossen, Joseph Jervase	#1571333159
15:40–16:00	**Dual-Layer Application-Identity Enforcement for Zero-Trust API Gateways in Microservices** Parman Sukarno, Aulia Arif Wardana, Farell Kresnanda	#1571341870
16:00–16:20	**AdaK-MoE: Entropy-Guided Adaptive Expert Budget for Efficient Mixture-of-Experts Inference** Yijun Lu, Zheng Lin, Zihan Fang, Yuyin Ma, Jing Yang, Lip Yee Por, Yinqiu Liu, Jialong Li	#1571330206
16:20–16:40	**A CNN-Aided PCG Detector for RIS-Aided OTFS Systems With Fractional Doppler Coupling** Jinlong Xie, Ke Mei, Siwen Dong, Jia Hou, Chaofeng Zhang	#1571326876
16:40–17:00	**Designing Cloud Caches over Hierarchical Storage** Lorenzo Marini, Damiano Carra	#1571334475

Friday 13 November

Paper sessions, BoF, Closing by 12:00, campus visit

08:30 **Registration** · Level 10

Parallel sessions · 09:00–10:20

PICOM-05 Wksp QuDiS 2026 Workshop

09:00–10:00 · 080.10.012 Online · Chair **Karine Even-Mendoza (King's College London, UK)**

09:00–09:15	**Cross-Layer Testing of Hybrid Quantum-Classical Distributed Applications: Fault Models, Testing Strategies, and Research Challenges** Vasileios Klimis	#1571339014
09:15–09:30	**Logic-Guided Hybrid Quantum SAT Solving with Prolog and Qiskit** Filippo Vella, Samuele Maria Morreale, Giovanni Pilato, Salvatore Gaglio	#1571346166
09:30–09:45	**Entropy as a Design Component for Quantum Systems** Tahereh Rezaei, Evan Hackertt, Kelle Clark, Peter Bordow	#1571350692
09:45–10:00	**A Gate Based Quantum Computing Framework for Codon Optimization** Fatemeh Ghasemi, Kion Kim	#1571345873

CST-24 WiP Work-in-Progress Papers

09:00–09:30 · 080.10.013 · Chair **Jianing Zhao (Waseda University, Japan)**

09:00–09:10	**MDR-GGCMF: A Multimodal Depression Recognition Framework with Global-Guided Cross Modal Fusion** WiP Jianing Zhao, Ruichen Cong, Qun Jin	#1571346519
09:10–09:20	**Rethinking AI Text Detection: Understanding Human, Social Bot, and ChatGPT Content Through Linguistic and Explainability Analysis** WiP Muhammad Owais, Kazuki Nakajima, Masaki Aida	#1571321878
09:20–09:30	**Attitude as the Mediating Mechanism in Auditors' Adoption of Artificial Intelligence: An Extended UTAUT Model** WiP Vigneswara Rao Subramaniam, Parvathy Rajendran, Yuvaraj Ganesan	#1571345527

CBD-09 Regular Tracks 2 & 3 (Americas, online) and Regular Paper

09:00–10:20 · 080.10.016 Hybrid · Chair **Lilan Yang (RMIT University, Australia)**

09:00–09:20	**What Does an Acknowledgment Guarantee? Durability and Performance in Replicated Log Brokers** online Raj Kumar Pandey, Pramod Kumar, Hardik Ruparel	#1571338218
09:20–09:40	**Bridging the Identity Gap: Purpose-Bound Delegation for MCP and A2A Agent Chain** online Khadarvali Shaik, Abdul Rasheed Shaik, Jyostna Seelam, Subramanyam Ramana, Syam Sajit, Venkata Phani Patelkhana	#1571341732
09:40–10:00	**Bounded Provisional Visibility: Controlling Poisoning Exposure in Continuously Ingested RAG Vector Stores** online Chuhong Xu, Lu Yi, Gangzhen Qian, Kainan Zhou, Zhaoyi Li, Hang Xiao	#1571334235
10:00–10:20	**Decentralized Identity Management for Multi-Cloud Environments Empowering SOLID PODs and Zero-Knowledge Authentication** Alfredo Cuzzocrea, Islam Belmerabet	#1571346037

10:20–10:40 *Morning Tea · Level 10*

10:40–11:40 **Birds-of-a-Feather (BoF)** · 080.01.002 · open to all four conferences

11:40–12:00 **Closing** · 080.01.002

12:00–13:30 *Lunch · Level 10*

13:30–15:30 **Visit RMIT AIDA Hub and Campus Tour**

15:30–16:00 *Afternoon Tea · Level 10*

16:00–17:00 **Committee Meeting**